

1) Free electrons:

$$H_e = \sum_k \epsilon_k a_k^\dagger a_k, \quad \epsilon_k = \frac{\hbar^2 k^2}{2m}$$

$$H_p = \sum_{\lambda} \omega_{\lambda} \left(b_{\lambda}^\dagger b_{\lambda} + \frac{1}{2} \right)$$

$$N_{\lambda} = \langle b_{\lambda}^\dagger b_{\lambda} \rangle = \frac{1}{e^{\omega_{\lambda}/T} - 1}$$

$$H_{e-p} = \sum_{\lambda, \sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) \underbrace{U(\vec{r} - \vec{R}_\lambda)}_{U(\vec{r} - \vec{R}_\lambda^0) - \vec{A}_\lambda \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\lambda^0)} \psi_\sigma(\vec{r})$$

$$H_{e-p} = - \sum_{\lambda, \sigma} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) \left(\vec{A}_\lambda \cdot \vec{\nabla} U(\vec{r} - \vec{R}_\lambda^0) \right) \psi_\sigma(\vec{r})$$

$$\vec{A}_\lambda = \frac{1}{\sqrt{N}} \sum_{\vec{q}} \sqrt{\frac{1}{2M\omega_{\lambda}}} \vec{\epsilon}_{\lambda} (b_{\lambda} + b_{-\lambda}^\dagger) e^{i\vec{q} \cdot \vec{R}_\lambda^0}$$

$$U = -\frac{ze^2}{r} e^{-\Lambda r}$$

$$\sum_{\vec{q}} e^{i(\vec{k} - \vec{k}' + \vec{q}) \cdot \vec{R}_\lambda^0} = N \sum_{\vec{q}} \delta_{\vec{k}, \vec{k}' + \vec{q} + \vec{G}}$$

$$- \sum_{\sigma, \lambda} \sum_{\vec{q}} \frac{1}{\sqrt{2M\omega_{\lambda} N}} \vec{\epsilon}_{\lambda} \frac{c_{\vec{k}\sigma}^\dagger c_{\vec{k}'\sigma}}{V} (b_{\lambda} + b_{-\lambda}^\dagger) e^{i\vec{q} \cdot \vec{R}_\lambda^0} \int d\vec{r} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} \underbrace{\vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{R}_\lambda^0|} e^{-\Lambda |\vec{r} - \vec{R}_\lambda^0|} \right)}_{\substack{\vec{r} - \vec{R}_\lambda^0 = \vec{x} \\ \Lambda \vec{r} = -i\vec{x}}}$$

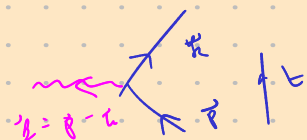
$$e^{i(\vec{q} + \vec{k} - \vec{k}') \cdot \vec{R}_\lambda^0} \int e^{i(\vec{k} - \vec{k}') \cdot \vec{x}} \vec{\nabla} U(\vec{x})$$

$$\sum_{\sigma, \lambda} \sum_{\vec{q}} \frac{i [\vec{\epsilon}_{\lambda} \cdot (\vec{k} - \vec{k}')] }{\sqrt{2M\omega_{\lambda}}} \frac{1}{V} \sqrt{N} U(\vec{k} - \vec{k}') \sum_{\vec{G}} \delta_{\vec{k}, \vec{k}' + \vec{q} + \vec{G}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}'\sigma} (b_{\lambda} + b_{-\lambda}^\dagger) =$$

$$= \frac{1}{\sqrt{V}} \sum_{\lambda, \vec{q}} g_{\lambda}(\vec{k}, \vec{q}) c_{\vec{k}+\vec{q}\sigma}^\dagger c_{\vec{k}\sigma} (b_{\lambda} + b_{-\lambda}^\dagger) ; \quad g_{\lambda}(\vec{k}, \vec{q}) = -i \frac{1}{\sqrt{2M\omega_{\lambda}}} \frac{4\pi ze^2}{(|\vec{q}|^2 + \Lambda^2)} \vec{q} \cdot \vec{\epsilon}_{\lambda}^*$$

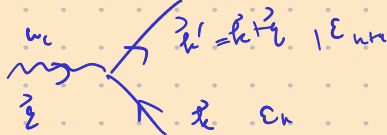
→ Fermi: spin ne drinja! $|\langle f | c_{\vec{k}}^\dagger c_{\vec{k}'} | i \rangle|^2 = n_{\vec{k}'} (1 - n_{\vec{k}})$

polu prazno

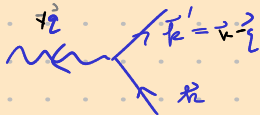


→ from trace: $K N_z - 1 |b_z| N_z \rangle|^2 = N_z$
 $| \langle N_z | b_z^\dagger | N_z + 1 \rangle |^2 = N_z + 1$

Fermi (variance): $\gamma_{i \rightarrow f} = 2\pi |K f | H_{fi} | i \rangle|^2 \underbrace{p(E_f)}_{\delta(E_f - E_i)}$

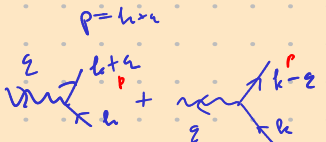
(1) Absorption process: 

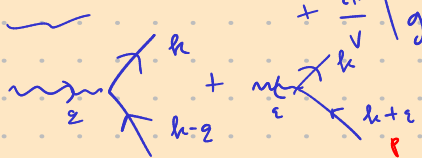
$|i\rangle = |N_z\rangle \otimes |k\rangle$
 $|f\rangle = |N_z+1\rangle \otimes |k+\vec{z}\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z+1 | (b_z + b_{-z}^\dagger) | N_z \rangle}_{N_z} \underbrace{\langle k+\vec{z} | c_{k+\vec{z}}^\dagger c_k | k \rangle}_{\sqrt{n_k (1-n_{k+\vec{z}})}} \cdot g \frac{1}{\sqrt{V}}$

(2) emission process: 

$|i\rangle = |N_z\rangle |k\rangle$
 $|f\rangle = |N_z+1\rangle |k-\vec{z}\rangle$
 $\rightarrow \sum_{k_z} \underbrace{\langle N_z+1 | b_z^\dagger | N_z \rangle}_{\sqrt{N_z+1}} \underbrace{\langle k-\vec{z} | c_{k-\vec{z}}^\dagger c_k | k \rangle}_{\sqrt{n_k (1-n_{k-\vec{z}})}} \cdot g \frac{1}{\sqrt{V}}$

(Mixture: $c_{k+\vec{z}}^\dagger c_k b_z + c_{k-\vec{z}}^\dagger c_k b_z^\dagger$)

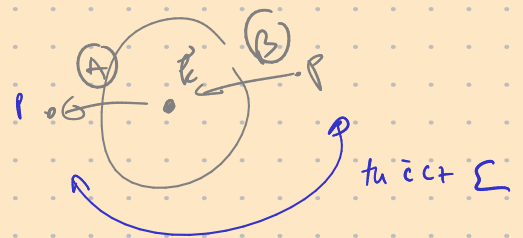
(A) $k \rightarrow p$: $\gamma_{k \rightarrow p} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1-n_{k+\vec{z}}) N_z \delta(\epsilon_{k+\vec{z}} - \epsilon_k - \omega_z) + \frac{2\pi}{V} |g_\lambda(\vec{k}, \vec{z})|^2 n_k (1-n_{k-\vec{z}}) (N_z+1) \delta(\epsilon_{k-\vec{z}} + \omega_z - \epsilon_k)$


(B) $p \rightarrow k$: $\gamma_{p \rightarrow k} = \gamma_{abs} + \gamma_{em} = \frac{2\pi}{V} |g_\lambda(\vec{p}, \vec{z})|^2 n_{k+\vec{z}} (1-n_k) (N_z+1) \delta(\epsilon_k + \omega_z - \epsilon_{k+\vec{z}}) + \frac{2\pi}{V} |g_\lambda(\vec{p}, \vec{z})|^2 n_{k-\vec{z}} (1-n_k) N_z \delta(\epsilon_k - \epsilon_{k-\vec{z}} - \omega_z)$


$N_{-z} = N_z$ for $\omega_z = \omega_{-z}$?

power in. strength

$I_k = \sum_p \gamma_{p \rightarrow k} - \gamma_{k \rightarrow p} = \dots$



$= \frac{2\pi}{V} \sum_{\vec{z}, \lambda} |g_\lambda(\vec{k}, \vec{z})|^2 \left\{ (1-n_k) n_{k+\vec{z}} (1+N_z) \delta(\epsilon_k - \epsilon_{k+\vec{z}} + \omega_z) + (1-n_k) n_{k-\vec{z}} N_z \delta(\epsilon_k - \epsilon_{k-\vec{z}} - \omega_z) - n_k (1-n_{k-\vec{z}}) (1+N_z) \delta(\epsilon_{k-\vec{z}} - \epsilon_k + \omega_z) - n_k (1-n_{k+\vec{z}}) N_z \delta(\epsilon_{k+\vec{z}} - \epsilon_k - \omega_z) \right\}$

① $q \rightarrow -q$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\hat{n}_k \rightarrow \hat{n}_{-k}$ $\delta(x-y) = \delta(y-x)$

$$\dot{n}_k = \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ (1-n_k) n_{k'} (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + \right. \\ \left. + (1-n_k) n_{k'} N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) (1+N_2) \delta(\epsilon_k + \epsilon_{k'} - \omega_2) - \right. \\ \left. - n_k (1-n_{k'}) N_2 \delta(\epsilon_k + \epsilon_{k'} + \omega_2) \right\} =$$

$$= \frac{2\pi}{V} \sum_{k'} |g_\lambda(k, k')|^2 \left\{ \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \left[(1-n_k) n_{k'} N_2 - (1-n_{k'}) n_k (1+N_2) \right] + \right. \\ \left. + \delta(\epsilon_k - \epsilon_{k'} + \omega_2) \left[(1-n_k) n_{k'} (1+N_2) - n_k (1-n_{k'}) N_2 \right] \right\} =$$

Bei demselben:

✓ analog hier (1.23b)

$$\dot{n}_k = \sum_{k'} \Gamma_{k' \rightarrow k}^{\lambda} (1-n_k) n_{k'} - \Gamma_{k \rightarrow k'}^{\lambda} (1-n_{k'}) n_k$$

$$\Gamma_{k' \rightarrow k}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_k - \epsilon_{k'} + \omega_2) + N_2 \delta(\epsilon_k - \epsilon_{k'} - \omega_2) \right\}$$

$$\Gamma_{k \rightarrow k'}^{\lambda} = \frac{2\pi}{V} |g_\lambda(k, k')|^2 \left\{ (1+N_2) \delta(\epsilon_{k'} - \epsilon_k + \omega_2) + N_2 \delta(\epsilon_{k'} - \epsilon_k - \omega_2) \right\}$$

2. način:

$$n_k = \text{Tr}[\rho_e \cdot c_k^\dagger c_k]$$

$$N_z = \text{Tr}[\rho_e \cdot b_z^\dagger b_z]$$

$$\rho_0 = \rho_e \otimes \rho_h$$

$$H_{eq} = \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) c_{k+q}^\dagger c_k (b_{2\lambda} + b_{-2\lambda}^\dagger) \quad ; \quad \rho \equiv k_1 \rho$$

$$c_k(t) = e^{-i\epsilon_k t} c_k \rightarrow c_k^\dagger c_k(t) = e^{i(\epsilon_k - \epsilon_k)t} c_k^\dagger c_k$$

$$b_{2\lambda}(t) = e^{-i\omega_{2\lambda} t} b_{2\lambda}$$

$$\rightarrow H_{eq}^I(t) = \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) c_{k+q}^\dagger c_k (b_{2\lambda} e^{i(\epsilon_k - \epsilon_{k+q} - \omega_{2\lambda})t} + b_{-2\lambda}^\dagger e^{i(\epsilon_k - \epsilon_{k+q} + \omega_{2\lambda})t})$$

$$\rightarrow I_k^I(t) = \dot{n}_k^I(t) = i [H_{eq}^I(t), n_k] \quad \text{in} \quad \text{Tr}[\rho_0 I_k^I] = 0, \quad \text{in} \quad I \propto \langle b_{2\lambda} + b_{-2\lambda}^\dagger \rangle = 0$$

- $\mathcal{O}(\omega^2)$:

$$\dot{n}_k = i^2 \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T dt \text{Tr}(\rho_0 [H_{eq}^I(t), [H_{eq}, n_k]])$$

$[c_k, n_k] = \delta_{kk} c_k$
 $[c_k^\dagger, n_k] = -\delta_{kk} c_k^\dagger$

$$(1) \quad \frac{1}{\sqrt{V}} \sum_{k,q,\lambda} g_\lambda(k,q) (b_{2\lambda} + b_{-2\lambda}^\dagger) \underbrace{[c_{k+q}^\dagger c_k, n_k]}_{c_{k+q}^\dagger c_k \delta_{kk} - c_k^\dagger c_{k+q} \delta_{k+q,k}} =$$

$$= \frac{1}{\sqrt{V}} \sum_{q,\lambda} g_\lambda^{(1)}(q) (b_{2\lambda} + b_{-2\lambda}^\dagger) (c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q})$$

$$(2) \quad \frac{1}{V} \sum_{k',q',\lambda'} \sum_{q,\lambda} g_{\lambda'}(k',q') g_\lambda(k,q) \left\{ (b_{2\lambda'}(t) + b_{-2\lambda'}^\dagger(t)) (b_{2\lambda} + b_{-2\lambda}^\dagger) [c_{k'+q'}^\dagger c_{k'}, c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q}] + \right. \\ \left. + [b_{2\lambda'}(t) + b_{-2\lambda'}^\dagger(t), b_{2\lambda} + b_{-2\lambda}^\dagger] \cdot c_{k'+q'}^\dagger c_{k'} (c_{k+q}^\dagger c_k - c_k^\dagger c_{k+q}) \right\}$$

... born kamije, pravi deke.

2. BCS currents za ph-el. coupling:

$$k = k_{\uparrow} \uparrow \quad -k = k_{\downarrow} \downarrow$$

$$H_e = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - \sum_k \left(\Delta c_k^\dagger c_{-k}^\dagger + \Delta^* c_k c_{-k} \right) \quad \begin{matrix} \xi_k = \epsilon_k - \mu \\ \xi_{-k} = \xi_k \end{matrix}$$

$$\rightarrow E_k = \sqrt{\xi_k^2 + \Delta^2}, \quad M_k = \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right), \quad N_k = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right), \quad M_k N_k = \frac{\Delta}{2E_k}$$

$$\rightarrow \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} M_k & N_k \\ -N_k & M_k \end{pmatrix} \begin{pmatrix} \gamma_k \\ \gamma_{-k}^\dagger \end{pmatrix} \Rightarrow H_e = E_0 + \sum_{k\sigma} E_k \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \quad N_{k\sigma} = \langle \gamma_{k\sigma}^\dagger \gamma_{k\sigma} \rangle$$

→ enaka interakcija: $H_{ep} = \frac{1}{V} \sum_{\vec{k}, \vec{k}', \sigma} g_{\lambda}(\vec{k}, \vec{k}') c_{\vec{k}+\vec{k}', \sigma}^\dagger c_{\vec{k}, \sigma} (b_{\vec{k}'}^\dagger + b_{-\vec{k}'}^\dagger)$

rotiramo

rotiramo s novo bazo: $(M_k' = M_{k+2}, N_k' = N_{k+2}) \quad (M_k' = M_{-k}, N_k' = N_{-k}!)$

$$\begin{aligned} c_{k\uparrow} &= M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger \\ c_{k\uparrow}^\dagger &= M_k \gamma_{k\uparrow}^\dagger + N_k \gamma_{-k\downarrow}^\dagger \end{aligned}$$

$$\begin{aligned} c_{k\downarrow} &= M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger \\ c_{k\downarrow}^\dagger &= M_k \gamma_{k\downarrow}^\dagger - N_k \gamma_{-k\uparrow}^\dagger \end{aligned}$$

$$\begin{aligned} \downarrow \\ \sum_{k\sigma} c_{k+\vec{q}, \sigma}^\dagger c_{k\sigma} &= \sum_k \left[(M_{k+2} \gamma_{k+2\uparrow}^\dagger + N_{k+2} \gamma_{-k-2\downarrow}^\dagger) (M_k \gamma_{k\uparrow} + N_k \gamma_{-k\downarrow}^\dagger) + (M_{k+2} \gamma_{k+2\downarrow}^\dagger - N_{k+2} \gamma_{-k-2\uparrow}^\dagger) (M_k \gamma_{k\downarrow} - N_k \gamma_{-k\uparrow}^\dagger) \right] \\ &= \sum_{k\sigma} M_{k+2} M_k (\gamma_{k+2\uparrow}^\dagger \gamma_{k\sigma}) + \sum_{k\sigma} N_{k+2} N_k (\gamma_{-k-2\downarrow}^\dagger \gamma_{-k\sigma}^\dagger) + \\ &\quad + \sum_k (M_{k+2} N_k \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger - N_{k+2} M_k \gamma_{k+2\downarrow}^\dagger \gamma_{-k\uparrow}^\dagger) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger \\ &\quad + \sum_k (N_{k+2} M_k \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow} - M_{k+2} N_k \gamma_{-k-2\uparrow}^\dagger \gamma_{k\downarrow}) \stackrel{h \rightarrow -q-h}{=} (M_{k+2} N_k + M_k N_{k+2}) \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow} \end{aligned}$$

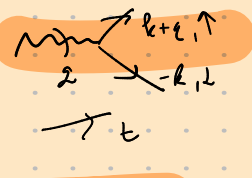
$$\begin{aligned} P_2^+ &= \sum_{k\sigma} c_{k\sigma}^\dagger c_{k\sigma} = 2 \delta_{\vec{q}, 0} \sum_k N_k^2 + \sum_{k\sigma} (M_{k+2} M_k - N_{k+2} N_k) \gamma_{k+2, \sigma}^\dagger \gamma_{k, \sigma} + \\ &\quad + \sum_k (M_{k+2} N_k + N_{k+2} M_k) [\gamma_{k+2\uparrow}^\dagger \gamma_{-k\downarrow}^\dagger + \gamma_{-k-2\downarrow}^\dagger \gamma_{k\uparrow}] \end{aligned}$$

scattering
ce ali ne

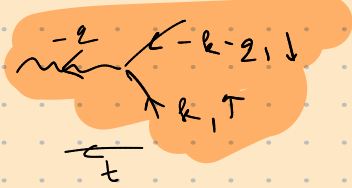
• scattering člena enaka, s faktorjem: $|M_{k+2} M_k - N_{k+2} N_k|^2$

$$\hookrightarrow g_{\lambda}(\vec{k}, \vec{k}') \rightarrow G_{\lambda}^s(\vec{k}, \vec{k}') = g_{\lambda} (M_{k+2} M_k - N_{k+2} N_k)$$

• pair creation: $H_{eg}^c = \frac{1}{V} \sum_{k,q,\lambda} \left[\underbrace{G_{\lambda}^c(k,q)}_{G_{\lambda}^c(k,q)} (n_{k+q,\uparrow} n_{k,\downarrow} + n_{k+q,\downarrow} n_{k,\uparrow}) \right] \gamma_{k+q,\uparrow}^{\dagger} \gamma_{k,\downarrow}^{\dagger} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$

$$|\langle f | H_{eg}^c | i \rangle|^2 \propto \underbrace{|N_{2-1} \langle b_{2\lambda} | N_{2-1} \rangle|^2}_{N_2} \cdot (1 - n_{k+q,\uparrow}) (1 - n_{k,\downarrow})$$


• pair annihilation: $H_{eg}^a \propto \gamma_{k+q,\downarrow} \gamma_{k,\uparrow} (b_{2\lambda} + b_{-2\lambda}^{\dagger})$



$$|\langle f | H_{eg}^a | i \rangle|^2 \propto \underbrace{|N_{2-1} \langle b_{2\lambda}^{\dagger} | N_{2-1} \rangle|^2}_{(N_{2-1}+1)} \cdot n_{k,\uparrow} n_{-k-q,\downarrow}$$

calculate for:

$$p = k + q$$

$$\dot{n}_k = \sum_p \Gamma_{p \rightarrow k}^s (1 - n_k) n_p - \Gamma_{k \rightarrow p}^s n_k (1 - n_p) + \Gamma_{k,p}^c (1 - n_{k+q,\uparrow}) (1 - n_{k,\downarrow}) - \Gamma_{k,p}^a n_{k,\uparrow} n_{-k-q,\downarrow}$$

$$\Gamma_{p \rightarrow k}^s = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^s(k,p)|^2 \cdot \left\{ (1 + N_2) \delta(E_k - E_p + \omega_2) + N_{-2} \delta(E_k - E_p - \omega_2) \right\}$$

$$\Gamma_{k \rightarrow p}^s = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^s(k,p)|^2 \cdot \left\{ (1 + N_{-2}) \delta(E_p - E_k + \omega_2) + N_2 \delta(E_p - E_k - \omega_2) \right\}$$

$$\Gamma_{k,p}^c = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^c(k,p)|^2 \cdot \left\{ N_2 \delta(E_p + E_k - \omega_2) \right\}$$

$$\Gamma_{k,p}^a = \frac{2\pi}{V} \sum_{\lambda} |G_{\lambda}^a(k,p)|^2 \cdot \left\{ (1 + N_{-2}) \delta(\omega_2 - E_p - E_k) \right\}$$

→ v. standard major.: $(u, v) \rightarrow (u, v)$ in (u, v) (majorize her pr. n. j. h. : $(\begin{smallmatrix} u & -v \\ v & u \end{smallmatrix})$, pr. men. $(\begin{smallmatrix} u & v \\ -v & u \end{smallmatrix})$.)

→ key je: $A_{\gamma-\delta}$ (namer $\delta(E_p - E_{p'} - \omega_2) \cdot \dots$??? $\delta(p - p' - \frac{\omega}{2})$?

3. EI currents, el. - ph.

$$H_e = \sum_k (c_{a,k}^\dagger, c_{b,k}^\dagger) \begin{pmatrix} \epsilon_a + v n_b & -\Delta^\dagger \\ -\Delta & \epsilon_b + v n_a \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix}$$

$$\hookrightarrow \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} m_k & r_k \\ -v_k & M_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}, \quad E_{\lambda k} = v_k \pm \sqrt{\xi_k^2 + \Delta^2};$$

$$\eta_k = \frac{\epsilon_a + \epsilon_b + v(n_a + n_b)}{2}, \quad \xi_k = \frac{\epsilon_a - \epsilon_b + v(n_a - n_b)}{2}$$

?

$$H_{ep} = \frac{1}{\sqrt{V}} \sum_{k, \lambda, \lambda'} (c_{a,k+\lambda}^\dagger, c_{b,k+\lambda}^\dagger) \begin{pmatrix} g_{a,\lambda}(k, \lambda) & 0 \\ 0 & g_{b,\lambda}(k, \lambda) \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} (b_{\lambda,1}^\dagger + b_{-\lambda,1}^\dagger)$$

$$\begin{aligned} c_{a,k+\lambda}^\dagger c_{a,k} &= (u_p \alpha_k^\dagger + v_p \beta_k^\dagger) (u_k \alpha_k + v_k \beta_k) = \\ &= u_p u_k \alpha_k^\dagger \alpha_k + v_p v_k \beta_k^\dagger \beta_k + u_p v_k \alpha_k^\dagger \beta_k + v_p u_k \beta_k^\dagger \alpha_k \end{aligned}$$

$$H_{ep} = \frac{1}{\sqrt{V}} \sum_{k, \lambda, \lambda'} (\alpha_{k+\lambda}^\dagger, \beta_{k+\lambda}^\dagger) \begin{pmatrix} g_a u_{k+\lambda} u_k + g_b v_{k+\lambda} v_k & g_a u_{k+\lambda} v_k - g_b v_{k+\lambda} u_k \\ g_a v_{k+\lambda} u_k - g_b u_{k+\lambda} v_k & g_a v_{k+\lambda} v_k + g_b u_{k+\lambda} u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} (b_{\lambda,1}^\dagger + b_{-\lambda,1}^\dagger)$$

$$G_{\mu\nu}(k, \lambda) = U_{\mu\lambda} (g_{\lambda\nu} \delta_{\lambda\nu}) U_{\nu\nu} \quad \begin{matrix} \lambda, \nu = a, b \\ \mu, \nu = a, b \end{matrix}$$

→ Fermi: $\mu, \nu = \{\alpha, \beta\}$

$$\Gamma_{\mu k \rightarrow \nu k} = 2\pi \sum_i |\langle f | H_{ep} | i \rangle|^2 \delta(E_f - E_i)$$

$$G_{\mu\nu} \hat{c}_{\mu, k+\lambda}^\dagger \hat{c}_{\nu, k} b_{\lambda,1}^\dagger = \text{diagram} \quad \frac{1}{V} |G_{\mu\nu}|^2 n_{\mu k} (1 - n_{\mu, k+\lambda}) N_{\lambda,1}$$

$$G_{\mu\nu} \hat{c}_{\mu, k+\lambda}^\dagger \hat{c}_{\nu, k} b_{-\lambda,1}^\dagger = \text{diagram} \quad \frac{1}{V} |G_{\mu\nu}|^2 n_{\mu k} (1 - n_{\mu, k+\lambda}) \cdot (1 + N_{\lambda,1})$$

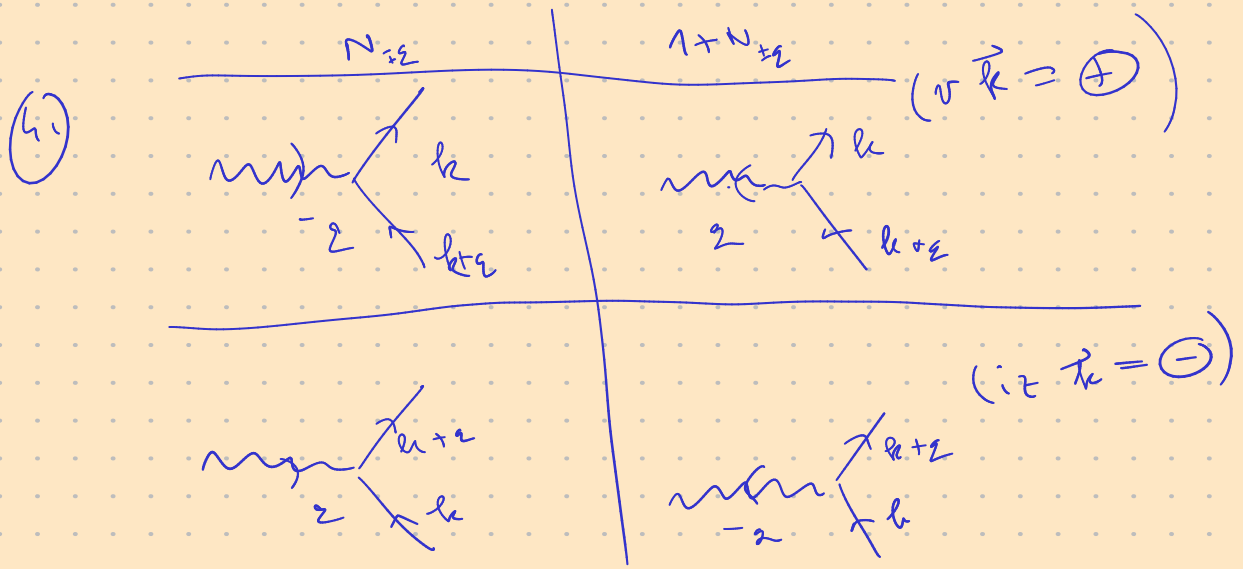
$\Gamma_{\nu p \rightarrow \mu k} = \text{diagram}$

$k \rightarrow (k) \xrightarrow{p}$

$$\dot{n}_{\alpha k} = \sum_p \left(\Gamma_{\alpha p \rightarrow \alpha k} n_{\alpha p} (1 - n_{\alpha k}) - \Gamma_{\alpha k \rightarrow \alpha p} n_{\alpha k} (1 - n_{\alpha p}) + \Gamma_{\beta p \rightarrow \alpha k} n_{\beta p} (1 - n_{\alpha k}) - \Gamma_{\alpha k \rightarrow \beta p} n_{\alpha k} (1 - n_{\beta p}) \right)$$

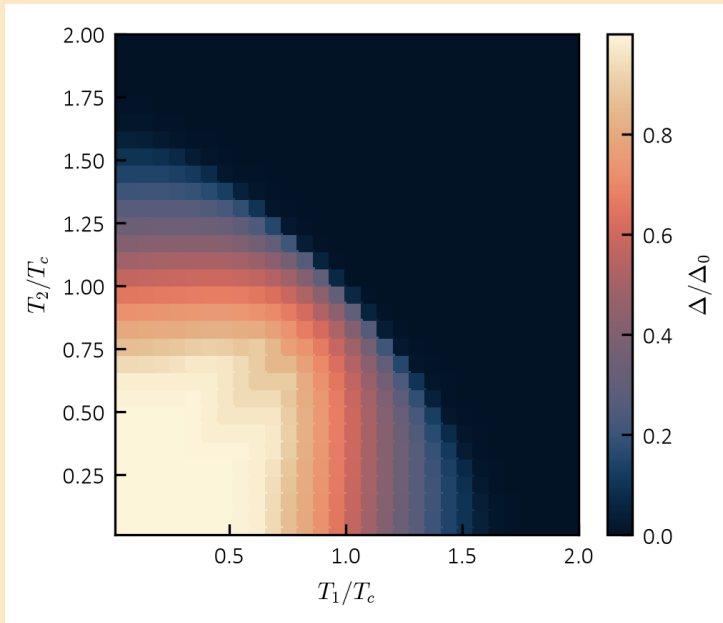
$$\Gamma_{\mu p \rightarrow \nu k} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^\lambda(k, \lambda)|^2 \left\{ (1 + N_{\lambda,1}) \delta(E_{\nu k} - E_{\mu p} + \omega_{\lambda,1}) + N_{\lambda,1} \delta(E_{\nu k} - E_{\mu p} - \omega_{\lambda,1}) \right\}$$

$$\Gamma_{\mu k \rightarrow \nu p} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^\lambda(k, \lambda)|^2 \left\{ (1 + N_{\lambda,1}) \delta(E_{\nu p} - E_{\mu k} + \omega_{\lambda,1}) + N_{\lambda,1} \delta(E_{\nu p} - E_{\mu k} - \omega_{\lambda,1}) \right\}$$



$$G^{\Lambda}(\vec{k}_{\pm 2}) = \begin{pmatrix} \mu_{k+z} & \nu_{k+z} \\ -\nu_{k+z} & \mu_{k+z} \end{pmatrix} \begin{pmatrix} g_{\Lambda 1}(\vec{q}) & 0 \\ 0 & g_{\Lambda 1}(\vec{q}) \end{pmatrix} \begin{pmatrix} \mu_k & \nu_k \\ -\nu_k & \mu_k \end{pmatrix}$$

$$g_{\Lambda 1}(\vec{q}) = -i \frac{\hbar \omega_{\vec{q}} e^2}{\sqrt{2M\omega_{\vec{q}}}} \cdot \frac{1}{|\vec{q}|^2 + \Lambda^2} (\vec{q} \cdot \vec{E}_{2\Lambda}^*)$$



4. free el. - photon int.

($e = -e$)

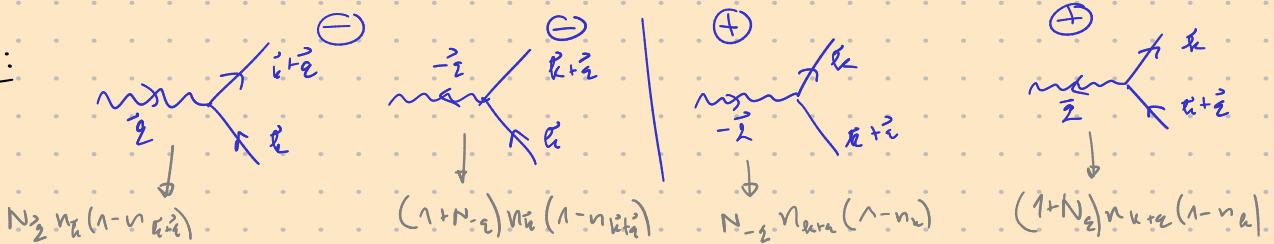
$$H = \frac{1}{2m} (\vec{p} + e\vec{A})^2 = \frac{\vec{p}^2}{2m} + \underbrace{\frac{e}{2m} \vec{p} \cdot \vec{A}}_{\substack{\text{"} \\ -(\vec{p} \cdot \vec{A}) - i\vec{A} \cdot \vec{p} = \vec{A} \cdot \vec{p}}} + \frac{e}{2m} \vec{A} \cdot \vec{p} + \frac{e^2}{2m} A^2 = \frac{\vec{p}^2}{2m} + \frac{e}{m} (\vec{A} \cdot \vec{p}) + \underbrace{\frac{e^2}{2m} A^2}_{\approx 0} \quad \mathcal{O}(A^2)$$

$$H = \sum_{\vec{k}\sigma} \epsilon_{\vec{k}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} + \frac{e}{m} \sum_{\vec{r}} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) [\vec{A}(\vec{r}) \cdot \vec{p}] \psi_\sigma(\vec{r})$$

$$\sum_{\vec{r}} \sum_{\vec{k}\vec{k}'} \frac{e}{m} \int d\vec{r} e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} c_{\vec{k}\sigma}^\dagger \left[\frac{1}{2\epsilon_0 V \omega_{\vec{k}}} \vec{\epsilon}_{\vec{k}} (a_{\vec{k}'} e^{i\vec{k}'\cdot\vec{r}} + a_{-\vec{k}'}^\dagger e^{i\vec{k}'\cdot\vec{r}}) \right] c_{\vec{k}\sigma} =$$

$$= \frac{1}{V} \sum_{\vec{r}} \sum_{\vec{k}\vec{k}'} \frac{e}{\sqrt{2\epsilon_0 \omega_{\vec{k}}}} (\vec{\epsilon}_{\vec{k}} \cdot (\vec{k} + \vec{k}')) \cdot c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} (a_{\vec{k}'} + a_{-\vec{k}'}^\dagger) = 11 e\gamma$$

→ Fermi:



$$\sum_{\vec{r}} T_{\vec{p} \rightarrow \vec{q}} = \sum_{\vec{r}, \vec{k}} \left(\frac{e^2}{2\epsilon_0 m^2 V \omega_{\vec{k}}} |\vec{\epsilon}_{\vec{k}} \cdot \vec{q}|^2 \right) \left\{ N_{-\vec{k}} \delta(\epsilon_k - \epsilon_p - \omega_{\vec{k}}) + (1 + N_{\vec{k}}) \delta(\epsilon_k - \epsilon_p + \omega_{\vec{k}}) \right\}$$

$$\sum_{\vec{r}} T_{\vec{q} \rightarrow \vec{p}} = \sum_{\vec{r}, \vec{k}} \left(\frac{e^2}{2\epsilon_0 m^2 V \omega_{\vec{k}}} |\vec{\epsilon}_{\vec{k}} \cdot \vec{p}|^2 \right) \left\{ N_{\vec{k}} \delta(\epsilon_q - \epsilon_k - \omega_{\vec{k}}) + (1 + N_{-\vec{k}}) \delta(\epsilon_q - \epsilon_k + \omega_{-\vec{k}}) \right\}$$

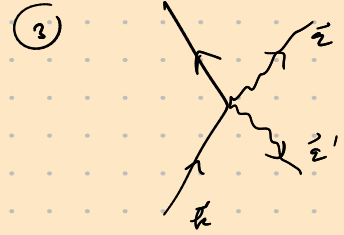
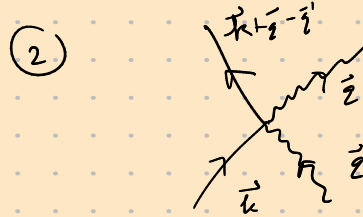
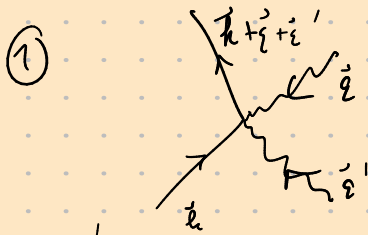
$$\mathcal{O}(A^2): H_{e\gamma}^{(2)} = \frac{e^2}{2m} \sum_{\vec{r}} \int d\vec{r} \psi_\sigma^\dagger(\vec{r}) (\vec{A}(\vec{r}) \cdot \vec{A}(\vec{r})) \psi_\sigma(\vec{r}) =$$

$$= \frac{e^2}{2m} \sum_{\vec{r}} \sum_{\vec{k}\vec{k}'} \int d\vec{r} e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} c_{\vec{k}\sigma}^\dagger \left[\sum_{\substack{\vec{q}\vec{q}' \\ \lambda\lambda'}} \frac{1}{2\epsilon_0 V} \frac{1}{\sqrt{\omega_{\vec{k}} \omega_{\vec{q}'}}} (\vec{\epsilon}_{\vec{k}} \cdot \vec{\epsilon}_{\vec{q}'}) \cdot (a_{\vec{k}} e^{i\vec{k}\cdot\vec{r}} + a_{-\vec{k}}^\dagger e^{i\vec{k}\cdot\vec{r}}) (a_{\vec{q}'} e^{i\vec{q}'\cdot\vec{r}} + a_{-\vec{q}'}^\dagger e^{i\vec{q}'\cdot\vec{r}}) \right] c_{\vec{k}\sigma}$$

$$\textcircled{1} \sum_{\substack{\vec{k}\vec{k}' \\ \vec{q}\vec{q}' \\ \lambda\lambda'}} \int d\vec{r} e^{i(\vec{k}-\vec{k}'+\vec{q}+\vec{q}')\cdot\vec{r}} \text{ or } c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} a_{\vec{q}} a_{\vec{q}'}$$

$$(a_{\vec{k}} a_{\vec{q}'} + a_{\vec{q}} a_{-\vec{q}'}^\dagger + a_{-\vec{k}}^\dagger a_{\vec{q}'} + a_{-\vec{q}}^\dagger a_{-\vec{q}'}^\dagger) e^{i(\vec{k}+\vec{q})\cdot\vec{r}} \sim \sum_{\vec{q}\vec{q}'} \delta_{\vec{k}, \vec{k}+\vec{q}-\vec{q}'}$$

$$H_{\text{er}}^{(2)} = \frac{e^2}{4 \epsilon_0 V m} \sum_{\substack{\vec{k}, \vec{q}, \vec{q}' \\ \sigma, \sigma', \lambda, \lambda'}} \frac{\vec{\epsilon}_{\vec{q}, \lambda} \cdot \vec{\epsilon}_{\vec{q}', \lambda'}}}{\omega_{\vec{q}} \omega_{\vec{q}'}} \cdot \left[\overset{(1)}{c_{\vec{k}+\vec{q}+\vec{q}', \sigma}^+ c_{\vec{k}, \sigma} a_{\vec{q}, \lambda} a_{\vec{q}', \lambda'}}} + \overset{(2)}{2 a_{\vec{q}, \lambda}^+ a_{\vec{q}', \lambda'}} c_{\vec{k}+\vec{q}-\vec{q}', \sigma}^+ c_{\vec{k}, \sigma}} + \right. \\ \left. + c_{\vec{k}-\vec{q}-\vec{q}', \sigma}^+ c_{\vec{k}, \sigma} a_{\vec{q}, \lambda}^+ a_{\vec{q}', \lambda'}}^+ \right]$$



→ Fermi: $\langle f | H_{\text{er}}^{(2)} | i \rangle^2$

① $\sqrt{n_k (1 - n_{k+q+q'})} N_{\vec{q}, \lambda} N_{\vec{q}', \lambda'}$

② $\sqrt{n_k (1 - n_{k+q+q'})} N_{\vec{q}} (N_{\vec{q}'} + 1)$

③ $\sqrt{n_k (1 - n_{k-q-q'})} (1 + N_{\vec{q}}) (1 + N_{\vec{q}'})$

$$\dot{n}_{k\sigma} = \sum_{\substack{\vec{q}, \vec{q}' \\ \lambda, \lambda'}} \frac{2\pi}{V} \left(\frac{e^2}{4 m \epsilon_0 V} \right)^2 \frac{(\vec{\epsilon}_{\vec{q}, \lambda} \cdot \vec{\epsilon}_{\vec{q}', \lambda'})}{\omega_{\vec{q}} \omega_{\vec{q}'}} N(\omega_{\vec{q}}) N(\omega_{\vec{q}'}) \cdot \delta(\epsilon_k - \epsilon_{k-q-q'} - \omega_{\vec{q}} - \omega_{\vec{q}'}) n_{k-q-q'} (1 - n_k) +$$

$$+ - \dots 2(1 + N_{\vec{q}}) N_{\vec{q}'} \delta \dots$$

$$+ (1 + N_{\vec{q}})(1 + N_{\vec{q}'}) - (k \leftrightarrow p)$$

Interaction picture formalism: $H = H_0 + H'$

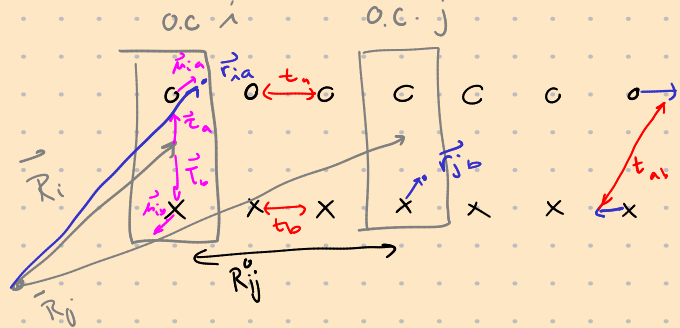
$$\begin{aligned}\frac{dH_{\text{int}}}{dt} &= \text{Tr}(n_{\text{int}} \dot{p}_{\pm}) = (-i)^2 \int_0^t dt' \text{Tr}\left(n_{\text{int}} \left[H'_I(t), [H'_I(t'), p_I(0)]\right]\right) = \\ &\quad \hookrightarrow \text{Tr}(A[B, C, D]) = \text{Tr}(D[A[B, C]]) \\ &= - \int_0^t dt' \langle [n_{\text{int}}, H'_I(t)], H'_I(t') \rangle\end{aligned}$$

$$H_e = \sum_j (E_a c_{ja}^\dagger c_{ja} + E_b c_{jb}^\dagger c_{jb}) - \sum_{\langle ij \rangle} \left(t_a(\vec{r}_j - \vec{r}_i) \cdot c_{ja}^\dagger c_{ia} + t_b(\vec{r}_j - \vec{r}_i) \cdot c_{jb}^\dagger c_{ib} + h.c. \right) +$$

da model ne da interband el.-ph int!

$$+ \sum_{\langle ij \rangle} t_{ab}(\vec{r}_j - \vec{r}_i) c_{aj}^\dagger c_{bi} + h.c.$$

$$\approx 0 + \vec{\nabla}_r t_{ab}(r) \cdot (\vec{u}_j - \vec{u}_i)$$



lahko (a_a, u_a, u_b, u_i)

$$\vec{d}_{ij,ba} = \vec{r}_{ja} - \vec{r}_{ib} = \underbrace{(\vec{r}_j - \vec{r}_i) + (\vec{r}_a - \vec{r}_b)}_{\vec{d}_{ij,ba}} + (\vec{u}_{ja} - \vec{u}_{ib})$$

$$t_a = t_a(d_{ij,aa}) \quad \text{in} \quad t_b = t_b(d_{ij,bb})$$

$$\Rightarrow t_a \approx t_a(d_{ij,ss}) + \frac{\partial}{\partial r} (t_a(r)) \bigg|_{d_{ij,ss}} (\vec{u}_{js} - \vec{u}_{is}) \quad R = (u, v)$$

→ assumc: Brskaj odzve se atom/orbitale premikajo skupaj: $\vec{r}_{ia} = \vec{u}_{ja} = \vec{u}_i$, etc. (in $\vec{r}_{ia} = \vec{u}_b = 0$)

$$① \quad t_a(d_{ijaa}) \approx t_a(r_{ij}) + \left(\vec{\nabla}_r t_a(r) \right) \cdot (\vec{u}_j - \vec{u}_i)$$

$$\approx \frac{\partial t_a(r)}{\partial r} \bigg|_{r_{ij}} \hat{R}_{ij} \equiv \alpha_a \hat{R}_{ij} \quad \vec{r} = (x, y)$$

$$② \quad t_{ab}(d_{ijab}) \approx 0 + \left(\vec{\nabla}_r t_{ab}(r) \right) \cdot (\vec{u}_j - \vec{u}_i) \approx \alpha_{ab} \cdot \hat{R}_{ij} \cdot (\vec{u}_j - \vec{u}_i)$$

②D

$$\rightarrow \vec{\nabla} [t_{ab}(x, y)] = \vec{\nabla} (xy \cdot f(r)) = (y, x) f(r) + xy \vec{\nabla} f(r) =$$

$\underbrace{(y, x)}_{\phi_a(r)\phi_b(r)} \quad \underbrace{f(r)}_{\alpha_{ab}} \quad \underbrace{xy \vec{\nabla} f(r)}_{\approx 0}$

$$H_e = \sum_i \epsilon_a^{(0)} c_{ia}^\dagger c_{ia} + \epsilon_b^{(0)} c_{ib}^\dagger c_{ib} - \sum_{\langle ij \rangle} (t_{ia} c_{ia}^\dagger c_{ja} + t_{ib} c_{ib}^\dagger c_{jb} + h.c.) + V \sum_j n_j - n_j \quad \text{with } V = \frac{W}{N}$$

$$H_{eg} = \sum_{\langle ij \rangle} \left(d_{ia} c_{ja}^\dagger c_{ia} \hat{R}_{ij}^0 + d_{ib} c_{jb}^\dagger c_{ib} \hat{R}_{ij}^0 + d_{ab} c_{ja}^\dagger c_{ib} + d_{ba} c_{ja}^\dagger c_{ib} + h.c. \right) \cdot (\vec{r}_j - \vec{r}_i)$$

→ first sum is: $V(\vec{r}_i - \vec{r}_j) \approx V \delta(\vec{r}_i - \vec{r}_j) + \mathcal{O}(|\vec{r}_j - \vec{r}_i|)$ & med utveckling se u. 2.10
 ↗ no order! coulomb-like!

$$\vec{r}_j = \frac{1}{\sqrt{N_{im}}} \sum_{\vec{z}, \lambda} \frac{1}{\sqrt{2M\omega_{\vec{z}}}} \hat{e}_{\vec{z}, \lambda} e^{i\vec{z} \cdot \vec{r}_j} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger)$$

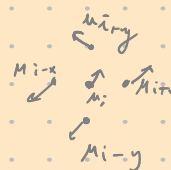
$$N_{im} = \frac{V}{v_{oc}}$$

$$c_{aj} = \frac{1}{\sqrt{N_{im}}} \sum_{\vec{k}} e^{i\vec{k} \cdot \vec{r}_j} c_{a\vec{k}}$$

→ new modification: $t_{ab} = 0$. compare: $\epsilon_a^{(0)} \rightarrow \epsilon_a^{(0)} + D_a \sum_{\langle ij \rangle} \hat{R}_{ij}^0 (\vec{r}_j - \vec{r}_i) + D_{ab} \sum_{\langle ij \rangle} \sigma_x \hat{R}_{ij}^0 (\vec{r}_j - \vec{r}_i)$

→ 2D model: $\hat{\delta} = (\delta_x, \delta_y) \Rightarrow \sum_{\langle ij \rangle} = \sum_{\vec{\delta}} \sum_{\vec{r}_i} (\delta \in \{\pm \vec{x}, \pm \vec{y}\})$

$$\vec{\delta} = (\delta_x, \delta_y) \Rightarrow \vec{r}_j - \vec{r}_i = \sum_{\vec{\delta}} \sum_{\vec{r}_i} (\vec{r}_i + \vec{\delta} - \vec{r}_i)$$



$$\begin{aligned} \sum_{\vec{\delta}} \hat{\delta} (\vec{r}_{i+\vec{\delta}} - \vec{r}_i) &= \hat{x} \cdot (\vec{r}_{i+x} - \vec{r}_i) - \hat{x} \cdot (\vec{r}_{i-x} - \vec{r}_i) + \hat{y} \cdot (\vec{r}_{i+y} - \vec{r}_i) - \hat{y} \cdot (\vec{r}_{i-y} - \vec{r}_i) \\ &= (n_{i+x}^x - n_{i-x}^x) + (n_{i+y}^y - n_{i-y}^y) \approx 2\vec{D} \cdot \vec{n} \end{aligned}$$

$$\sum_{\vec{\delta}} \tilde{\delta} (\vec{r}_{i+\vec{\delta}} - \vec{r}_i) = \tilde{y} \cdot (\vec{r}_{i+x} - \vec{r}_i) - \tilde{y} \cdot (\vec{r}_{i-x} - \vec{r}_i) + \dots = (n_{i+x}^y - n_{i-x}^y) - (n_{i+y}^x - n_{i-y}^x)$$

is then defined:

varianella ispeljivna

semo interband

$$H_{e-p} = \sum_{i, \vec{\delta}} \left(D_a c_{ai}^\dagger c_{ai} + D_b c_{bi}^\dagger c_{bi} + d_a c_{ai}^\dagger c_{i+\vec{\delta}} + d_b c_{bi}^\dagger c_{i+\vec{\delta}} \right) \cdot \hat{\delta} (\vec{r}_{i+\vec{\delta}} - \vec{r}_i) +$$

$$+ \sum_{i, \vec{\delta}} \left(D_{ab} c_{ai}^\dagger c_{bi} + h.c. \right) \tilde{\delta} \cdot (\vec{r}_{i+\vec{\delta}} - \vec{r}_i) \quad \leftarrow \text{da interband coupling}$$

$$\vec{r}_{i+\vec{\delta}} - \vec{r}_i = \frac{1}{N} \sum_{\vec{z}, \lambda} \frac{1}{\sqrt{2M\omega_{\vec{z}}}} \hat{e}_{\vec{z}, \lambda} (e^{i\vec{z} \cdot \vec{\delta}} - 1) e^{i\vec{z} \cdot \vec{r}_i} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^\dagger), \quad \hat{e}_{\vec{z}, \lambda} = \vec{e}_{-\vec{z}, \lambda}^*$$



$$\sum_{i \in \vec{\delta}} \delta(\vec{r}_{i,0} - \vec{r}_i) c_{\alpha i}^+ c_{\beta i+s} = \sum_{\vec{z}, \lambda} \frac{(b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)}{\sqrt{2NM\omega_{\vec{z}, \lambda}}} \sum_{\vec{\delta}} \hat{\vec{\delta}} \cdot \vec{\epsilon}_{\vec{z}, \lambda} (e^{i\vec{z}\vec{\delta}} - 1) \frac{1}{N} \sum_{\vec{r}} e^{i\vec{r}\vec{s}} c_{\alpha \vec{r}}^+ c_{\beta \vec{r}} \sum_i e^{i(\vec{z} + \vec{r} - \vec{r}_i) \cdot \vec{r}_i} =$$

$\delta_{\vec{z} + \vec{r}, \vec{r}} \pmod{\vec{L}}$

$$= \frac{1}{\sqrt{N}} \sum_{\vec{z}, \lambda} \left[\frac{1}{\sqrt{2M\omega_{\vec{z}, \lambda}}} \sum_{\vec{\delta}} \hat{\vec{\delta}} \cdot \vec{\epsilon}_{\vec{z}, \lambda} (e^{i\vec{z}\vec{\delta}} - 1) e^{i\vec{r}\vec{s}} \right] c_{\alpha \vec{r} + \vec{z}}^+ c_{\beta \vec{r}} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)$$

$$\alpha, \beta \in \{1, 2\}$$

$$\vec{z} \cdot (-\vec{x}) = -z_x$$

$$\begin{aligned} \rightarrow \sum_{\vec{\delta} = \{\pm \vec{x}, \pm \vec{y}\}} (e^{i\vec{z}\vec{\delta}} - 1) e^{i\vec{r}\vec{s}} \hat{\vec{\delta}} \cdot \vec{\epsilon}_{\vec{z}, \lambda} &= \epsilon_{\vec{z}, \lambda}^x \left[(e^{i z_x} - 1) - (e^{-i z_x} - 1) \right] e^{i k_x} \\ &+ \epsilon_{\vec{z}, \lambda}^y \left[(e^{i z_y} - 1) - (e^{-i z_y} - 1) \right] e^{i k_y} \\ &= 2i \left(\sin(k_x + z_x) - \sin(k_x) \right) \end{aligned}$$

$$\Rightarrow H_{ep} = \frac{1}{\sqrt{N}} \sum_{\vec{r}, \vec{z}, \lambda} (c_{\alpha \vec{r} + \vec{z}}^+, c_{\beta \vec{r} + \vec{z}}^+) \begin{pmatrix} g_{aa}^{\vec{\lambda}}(t, \vec{z}) & g_{ab}^{\vec{\lambda}} \\ g_{ba}^{\vec{\lambda}} & g_{bb}^{\vec{\lambda}} \end{pmatrix} \begin{pmatrix} c_{\alpha \vec{r}} \\ c_{\beta \vec{r}} \end{pmatrix} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)$$

$$g_{aa}^{\vec{\lambda}}(t, \vec{z}) = \frac{2i}{\sqrt{2M}} \frac{1}{\sqrt{\omega_{\vec{z}, \lambda}}} \left[\epsilon_{\vec{z}, \lambda}^x \left(D_a \sin(q_x) + \alpha_a \left(\sin(k_x + z_x) - \sin(k_x) \right) \right) + \epsilon_{\vec{z}, \lambda}^y (x \rightarrow y) \right]$$

$\alpha_a \propto \sqrt{\omega_{\vec{z}, \lambda}^2} - \omega(\vec{z})$

$$g_{ab}^{\vec{\lambda}}(t, \vec{z}) = \frac{2i}{\sqrt{2M}} \frac{1}{\sqrt{\omega_{\vec{z}, \lambda}}} \left[\epsilon_{\vec{z}, \lambda}^y (D_{ab} \sin(q_x)) + \epsilon_{\vec{z}, \lambda}^x (D_{ab} \sin(q_y)) \right]$$

$$H_{ep}^D = \frac{1}{\sqrt{N}} \sum_{\vec{r}, \vec{z}, \lambda} (\alpha_{\vec{r} + \vec{z}}^+, \beta_{\vec{r} + \vec{z}}^+) \underbrace{\begin{pmatrix} \mu_{\vec{r} + \vec{z}} & -\nu_{\vec{r} + \vec{z}} \\ \nu_{\vec{r} + \vec{z}} & \mu_{\vec{r} + \vec{z}} \end{pmatrix}}_{\underline{G}(\vec{r}, \vec{z})} \underline{g}(\vec{r}, \vec{z}) \begin{pmatrix} \mu_{\vec{r}} & \nu_{\vec{r}} \\ -\nu_{\vec{r}} & \mu_{\vec{r}} \end{pmatrix} \begin{pmatrix} \alpha_{\vec{r}} \\ \beta_{\vec{r}} \end{pmatrix} (b_{\vec{z}, \lambda} + b_{-\vec{z}, \lambda}^+)$$

$\rightarrow$ a to symmetric order: force atom zero-energy / non minimum ($D_{ab} \neq 0$)

T.b. el. - form hamiltonke:

$$p(\vec{r}) = -e \psi^\dagger(\vec{r}) \psi(\vec{r})$$

$$\psi(\vec{r}) = \sum_{i\alpha} \phi_\alpha(\vec{r}-\vec{r}_i) c_{i\alpha} \quad \bigg| \int d\vec{r} e^{i\vec{z}\cdot\vec{r}}$$

$$p(\vec{z}) = -e \sum_{ij} \sum_{\alpha\beta} \int d\vec{r} e^{-i\vec{z}\cdot\vec{r}} \phi_\alpha^\dagger(\vec{r}-\vec{r}_i) \phi_\beta(\vec{r}-\vec{r}_j) = -e \sum_{ij} \sum_{\alpha\beta} e^{i\vec{z}\cdot\vec{r}_i} \langle \phi_\alpha | e^{i\vec{z}\cdot\vec{x}} | \phi_\beta \rangle c_{i\alpha}^\dagger c_{j\beta} \delta_{ij}$$

$\vec{r} = \vec{r}_i + \vec{x}$

$$-i \frac{\vec{z}}{2} \cdot \langle \phi_\alpha | \vec{x} | \phi_\alpha \rangle c_{i\alpha}^\dagger c_{i\alpha} - i \frac{\vec{z}}{2} \cdot \langle \phi_i | \vec{x} | \phi_i \rangle c_{i\alpha}^\dagger c_{i\alpha} = 1 - i \vec{z} \cdot \vec{x}$$

$(c_{i\alpha}^\dagger c_{i\alpha})^\dagger = c_{i\alpha}^\dagger c_{i\alpha}$

$$\rightarrow p(\vec{z}) \approx -e \sum_i e^{-i\vec{z}\cdot\vec{r}_i} \left(n_{i\alpha} + n_{i\beta} - i \left(\frac{\vec{z}}{2} \cdot \langle \phi_\alpha | \vec{x} | \phi_\alpha \rangle \right) c_{i\alpha}^\dagger c_{i\alpha} + h.c. \right)$$

$-i\alpha - i\alpha^* = -i\alpha + (i\alpha)^*$

$$\rightarrow \sum_i e^{i\vec{z}\cdot\vec{r}_i} c_{i\alpha}^\dagger c_{i\beta} = \frac{1}{N} \sum_k c_{a,k+\vec{z}}^\dagger c_{b,k}$$

$\frac{1}{\sqrt{N}} \sum_k c_{a,k}^\dagger e^{-ik\cdot\vec{r}_i}$

assume same a, b dipol.

$$p(-\vec{z}) = -e \sum_k (c_{a,k+\vec{z}}^\dagger, c_{b,k+\vec{z}}^\dagger) \left(1 + \vec{z} \cdot \vec{d} \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right) \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix}$$

$$\rightarrow H_{ex} = - \int d\vec{r} \vec{j}_e \cdot \vec{A} = - \int d\vec{r} \sum_{\vec{z}} \vec{A}_{\vec{z}} e^{i\vec{z}\cdot\vec{r}} \cdot \vec{j}_e(\vec{r}) = - \sum_{\vec{z}} \vec{A}_{\vec{z}} \cdot \left(\int d\vec{r} e^{i\vec{z}\cdot\vec{r}} \vec{j}_e(\vec{r}) \right) = - \sum_{\vec{z}} \vec{j}_e(-\vec{z}) \cdot \vec{A}(\vec{z})$$

$\vec{j}_e(\vec{z}) \quad (\pm)?$

$$\vec{j}_e(\vec{r}) = - \frac{e}{m} \left(\psi^\dagger(\vec{r}) (-i \vec{\nabla}) \psi(\vec{r}) \right)$$

to get the fac. el.

$$-i \left[\int e^{-i\vec{z}\cdot\vec{r}} H_e(\vec{r}) d\vec{r} p(\vec{z}) \right]$$

→ to the TB model:

$$\text{Heisenberg: } \frac{\partial p(\vec{r})}{\partial t} = i [H_e, p] \quad \left(\dot{m} \right) \quad \frac{\partial p}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$$

$$H_e = \sum_i c_a^{(i)\dagger} c_a c_a + c_b^{(i)\dagger} c_b c_b - \sum_{(ij)} (t_{ia} c_a^\dagger c_a c_{ja} + t_{ib} c_b^\dagger c_b c_{jb} + h.c.) + V \sum_j n_j$$

$$i \vec{z} \cdot \vec{j}(\vec{z}) = i [H_e, p(\vec{z})]$$

? a V na uprelevaro?

$$H_e' = \sum_{\vec{r} \in \{a,b\}} \epsilon_{\vec{r},k} c_{\vec{r},k}^\dagger c_{\vec{r},k}$$

$$-e \sum_{k,k'} \left[\psi_k^\dagger \begin{pmatrix} c_{a,k} & 0 \\ 0 & c_{b,k} \end{pmatrix} \psi_{k'} + \psi_{k-\vec{z}}^\dagger \begin{pmatrix} 1 & -i \vec{z} \cdot \vec{d} \\ i \vec{z} \cdot \vec{d} & 1 \end{pmatrix} \psi_{k'} \right] =$$

$$\rightarrow \text{identitiki: } [\psi_k^\dagger A \psi_k, \psi_{k'-\vec{z}}^\dagger B \psi_{k'}] = \psi_{k'-\vec{z}}^\dagger (B A(k') - A(k'-\vec{z}) B) \psi_{k'} \delta_{k,k'+\vec{z}} = \psi_k^\dagger (B A(k+\vec{z}) - A(k) B) \psi_{k+\vec{z}}$$

$$= -e \sum_{\vec{k}} \psi_{\vec{k}}^{\dagger} \left[\begin{pmatrix} \epsilon_{a, \vec{k}+\vec{q}} & -i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{b, \vec{k}+\vec{q}} \\ +i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{a, \vec{k}} & \epsilon_{b, \vec{k}+\vec{q}} \end{pmatrix} - \begin{pmatrix} \epsilon_{a, \vec{k}} & -i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{b, \vec{k}} \\ +i(\vec{q} \cdot \vec{d}) \cdot \epsilon_{a, \vec{k}} & \epsilon_{b, \vec{k}} \end{pmatrix} \right] \psi_{\vec{k}+\vec{q}}$$

→ here follow: $\vec{q} \approx 0 \rightarrow \epsilon_{a, \vec{k}+\vec{q}} \sim \epsilon_{a, \vec{k}} + \vec{q} \cdot \nabla_{\vec{k}} \epsilon_{a, \vec{k}} \quad (*)$

→ $\epsilon_{a, \vec{k}+\vec{q}} - \epsilon_{a, \vec{k}} \approx \vec{q} \cdot \nabla_{\vec{k}} \epsilon_{a, \vec{k}}$

→ $\vec{q} \cdot \vec{j}(\vec{q}) \approx \vec{q} \cdot \left(-e \sum_{\vec{k}} \psi_{\vec{k}}^{\dagger} \begin{pmatrix} \nabla_{\vec{k}} \epsilon_{a, \vec{k}+\vec{q}} & -i(\epsilon_{b, \vec{k}+\vec{q}} - \epsilon_{a, \vec{k}}) \cdot \vec{d} \\ +i\vec{d}(\epsilon_{a, \vec{k}+\vec{q}} - \epsilon_{b, \vec{k}}) & \nabla_{\vec{k}} \epsilon_{b, \vec{k}+\vec{q}} \end{pmatrix} \psi_{\vec{k}} \right) \Rightarrow$

→ $\vec{j}(\vec{q}) = -e \sum_{\vec{k}} \psi_{\vec{k}-\vec{q}}^{\dagger} \begin{pmatrix} \nabla_{\vec{k}} \epsilon_{a, \vec{k}-\vec{q}} & +i\vec{d}(\epsilon_{a, \vec{k}-\vec{q}} - \epsilon_{b, \vec{k}}) \\ -i\vec{d}(\epsilon_{b, \vec{k}-\vec{q}} - \epsilon_{a, \vec{k}}) & \nabla_{\vec{k}} \epsilon_{b, \vec{k}-\vec{q}} \end{pmatrix} \psi_{\vec{k}} \quad (+)$

(zaključiti $\oplus$ primer za free el. $\rightarrow \frac{(\hbar v)^2}{2m} - \frac{\hbar^2}{2m} = \frac{1}{2m} (2\vec{q} \cdot \vec{k} + q^2) = \vec{q} \cdot \frac{1}{2m} (2\vec{k} + \vec{q}) = \vec{q} \cdot \left(\frac{\vec{k} + \frac{\vec{q}}{2}}{m} \right)$)

+ b. $\rightarrow -\cos(k_q) + \cos(k) = 2 \sin \frac{q}{2} \sin \left(k + \frac{q}{2} \right) \approx 2 \cdot \frac{\sin \left(k + \frac{q}{2} \right)}{\nabla_{\vec{k}} \epsilon_{k+\frac{q}{2}}}$

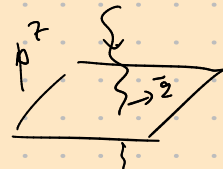
$H_{eg} = -\sum_{\vec{q}} \vec{j}(\vec{q}) \cdot \vec{A}(\vec{q}) = \sum_{\vec{q}} \frac{-e}{\sqrt{2V\epsilon_0\omega_{\vec{q}}}} \psi_{\vec{k}-\vec{q}}^{\dagger} \begin{pmatrix} \vec{\epsilon}_{\vec{q}} \cdot \vec{p}_a(\vec{k}+\frac{\vec{q}}{2}) & +i(\vec{\epsilon}_{\vec{q}} \cdot \vec{d})(\epsilon_{a, \vec{k}+\frac{\vec{q}}{2}} - \epsilon_{b, \vec{k}}) \\ -i(\vec{\epsilon}_{\vec{q}} \cdot \vec{d})(\epsilon_{b, \vec{k}-\frac{\vec{q}}{2}} - \epsilon_{a, \vec{k}}) & \vec{\epsilon}_{\vec{q}} \cdot \vec{p}_b(\vec{k}+\frac{\vec{q}}{2}) \end{pmatrix} \psi_{\vec{k}} (a_{\vec{q}, \lambda} + a_{-\vec{q}, \lambda}^{\dagger})$

more like $\ominus \oplus$!

$\underline{M^A(\vec{k}, \vec{q})}$

$\vec{k} - \vec{q} \rightarrow \vec{k} + \vec{q}$ (nekeje je narobe, preveri kje si jelo minus.)

→ simple polarizacije: setup:



$\vec{q} = q(\cos \varphi, \sin \varphi)$

$\vec{E}_{TE}(z) = (\hat{z} \times \vec{q}) \sin\left(\frac{\pi}{L} z\right)$

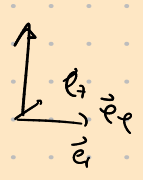
$\vec{E}_{TM}(z) = \frac{\omega_0}{\omega_z} \hat{z} \sin\left(\frac{\pi}{L} z\right) + i \frac{c z}{\omega_z} \hat{z} \cos\left(\frac{\pi}{L} z\right) \rightarrow z = \frac{L}{2} \Rightarrow$

$\vec{E}_{TE} \approx (-\sin \varphi, \cos \varphi)$
 $\vec{E}_{TM} \approx \frac{\omega_0}{\omega_z} (\cos \varphi, \sin \varphi)$

$\vec{p}_{a,b}(\vec{k} + \frac{\vec{q}}{2}) = 2\hbar a_{1,2} \left(\sin\left(k_x + \frac{q_x}{2}\right), \sin\left(k_y + \frac{q_y}{2}\right) \right), \quad \vec{d} = \begin{cases} d(\cos \theta, \sin \theta) & \text{in plane} \\ d\hat{z} & \text{out of plane} \end{cases}$

	TE	TM
$\vec{E} \cdot \vec{r}_\perp (k_\perp \vec{z})$	$2 \text{tr} \left(-\sin \gamma \sin \left(k_x + \frac{2x}{L} \right) + \cos \gamma \sin \left(k_y + \frac{2y}{L} \right) \right)$	$\frac{\omega_0}{\omega_z} 2 \text{tr} \left(\cos \gamma \sin \left(k_x + \frac{2x}{L} \right) + \sin \gamma \sin \left(k_y + \frac{2y}{L} \right) \right)$
$\vec{E} \cdot \vec{d}$ in plane	$d \sin(\theta - \varphi)$ ni k ali q dependent.	$\frac{\omega_0}{\omega_z} d \cos(\theta - \varphi)$
$\vec{E} \cdot \vec{d}$ canon axis	0	0 pri $z = \frac{L}{2}$.

$$\vec{d} = \langle \phi_a | \vec{r} | \phi_b \rangle = \int \phi_a^*(\vec{r} - \vec{r}_i) \vec{r} \phi_b(\vec{r} - \vec{r}_i) d\vec{r} \quad / \cdot \hat{e}_x$$

$$d_y^{(1)} = d_x = \int \phi_a^*(\vec{r} - \vec{r}_i) r \cos \theta \phi_b(\vec{r} - \vec{r}_i) d\vec{r} \dots$$


$$\rightarrow \text{relevantno je } \sum_{\lambda} |\vec{C}_\lambda \cdot \vec{X}|^2 = |\hat{z} \times \vec{X}|^2 + \frac{\omega_z^2}{\omega_0^2} |\hat{z} \cdot \vec{X}|^2 \xrightarrow{\vec{z} \rightarrow 0} |\vec{X}|^2$$

$$\rightarrow \text{vredniti: } 4 \text{tr}^2 (\sin^2 k_x + \sin^2 k_y), \text{ mi odvisen od realnih koordinat!}$$

Integriramo: resitev enačbe:

$$\vec{z} \cdot \vec{j}(\vec{z}) = \vec{f}(\vec{z}, t)$$

$$\vec{a} \cdot \vec{x} = f(\vec{a}, t) \quad / \cdot \vec{a}$$

$$(\vec{a} \cdot \vec{a}) \cdot \vec{x} = \vec{a} \cdot \vec{f} \rightarrow \vec{j}(\vec{z}) = \frac{1}{z^2} (\vec{z} \cdot \vec{f}(\vec{z}, t))$$

$$H_e^{(0)} = \sum_{i,j} \epsilon_{ij}^0 c_{ji}^\dagger c_{ji} + \sum_{\langle ij \rangle} t_{ij} c_{ij}^\dagger c_{ji}$$

$$\psi(\vec{r}) = \sum_{i,j} \phi_{ij}(\vec{r} - \vec{r}_{ij}) c_{ji}$$

$$H_e^{(0)} = \int \psi^\dagger(\vec{r}) \psi(\vec{r}) d\vec{r}$$

$$\int \sum_{i,j} \phi_{ij}^*(\vec{r} - \vec{r}_{ij}) \phi_{ij}(\vec{r} - \vec{r}_{ij}) d\vec{r} c_{ji}^\dagger c_{ji}$$

$$\int \phi_{ij}^*(\vec{r} - \vec{r}_{ij}) \phi_{ij}(\vec{r} - \vec{r}_{ij}) d\vec{r} = \epsilon_{ij}^0$$

$$t_{ij}(\vec{r}) \quad j = t_{ij}(\vec{r})$$

$$H_e^{(0)} = \int e^{i\vec{z} \cdot \vec{r}} d\vec{r} \psi^\dagger(\vec{r}) \psi(\vec{r}) =$$

$$\epsilon_{k_2} - \epsilon_{k_1} = \frac{k^2}{2m} - \frac{(k+q)^2}{2m} = \frac{1}{2m} (k^2 - (k^2 + 2\vec{k} \cdot \vec{q} + q^2)) =$$

$$= -\frac{1}{2m} (2\vec{k} \cdot \vec{q} + q^2) \sim -\frac{1}{2m} \epsilon_{k_1} \epsilon_{k_2}$$

$$\epsilon_{k_2} - \epsilon_{k_1} = \frac{(k_2 - q)^2 - k^2}{2m} = \frac{k^2 - 2qk + q^2 - k^2}{2m} = -\frac{2qk - q^2}{2m} \sim -\frac{2qk}{2m} \sim -\frac{1}{2m} \epsilon_{k_1} \epsilon_{k_2}$$

Electron-photon coupling zu free el.

$$\psi(\vec{r}) = \frac{1}{\sqrt{V}} \sum_{\vec{k}} e^{i\vec{k} \cdot \vec{r}} c_{\vec{k}}$$

$$H_e = \sum_{\vec{k}} \underbrace{(\epsilon_0 + \hbar k^2)}_{\epsilon_{\vec{k},1}} + H_v$$

$$= \sum_{\vec{k} \in \{a,1\}} \epsilon_{\vec{k},1} + H_v$$

$$\pm \frac{G}{2} \mp 2t(\cos k_x + \cos k_y)z$$

$$k_x \sim k_y \sim 0$$

$$= \pm \frac{G}{2} \mp 2t \left(1 - \frac{k_x^2}{2} + 1 - \frac{k_y^2}{2} \right) =$$

$$= \left(\pm \frac{G}{2} \mp 4t \right) \pm t(k_x^2 + k_y^2) = \pm \left(\epsilon^0 + \hbar \vec{k} \cdot \vec{k} \right)$$

$$\epsilon^0 = \frac{G}{2} - 4t$$

$$H_{cr}^{(1)} = - \int \vec{j}_e \cdot \vec{A} d\vec{r} = - \sum_{\vec{k}} \vec{j}_e(-\frac{\vec{k}}{2}) A(\frac{\vec{k}}{2})$$

$$\vec{j}_e(\vec{r}) = -\frac{e}{2m} (\psi^\dagger(\vec{r}) \vec{\nabla} \psi(\vec{r}) - (\vec{\nabla} \psi^\dagger(\vec{r})) \psi(\vec{r})) = -\frac{e}{2m} \frac{1}{V} \sum_{\vec{k}, \vec{k}'} \left(e^{-i\vec{k}' \cdot \vec{r}} e^{i\vec{k} \cdot \vec{r}} i\vec{k} c_{\vec{k}'}^\dagger c_{\vec{k}} - e^{i\vec{k}' \cdot \vec{r}} e^{-i\vec{k} \cdot \vec{r}} (-i\vec{k}') c_{\vec{k}'}^\dagger c_{\vec{k}} \right)$$

$$\vec{j}_e(-\frac{\vec{k}}{2}) = \int d\vec{r} e^{i\frac{\vec{k}}{2} \cdot \vec{r}} \vec{j}_e(\vec{r}) = -\frac{e}{2m} \sum_{\vec{k}, \vec{k}'} \underbrace{\left(\frac{1}{V} \int d\vec{r} e^{i(\frac{\vec{k}}{2} + \vec{k} - \vec{k}') \cdot \vec{r}} \right)}_{\delta_{\vec{k}', \vec{k} + \frac{\vec{k}}{2}}} c_{\vec{k}'}^\dagger c_{\vec{k}} \vec{k} + \underbrace{\left(\frac{1}{V} \int d\vec{r} e^{i(\vec{k} + \vec{k} - \vec{k}') \cdot \vec{r}} \right)}_{\delta_{\vec{k}', \vec{k} + \frac{\vec{k}}{2}}} c_{\vec{k}'}^\dagger c_{\vec{k}} \vec{k} =$$

$$= -\frac{e}{2m} \sum_{\vec{k}} \left(c_{\vec{k} + \frac{\vec{k}}{2}}^\dagger c_{\vec{k}} \vec{k} + (\vec{k} + \frac{\vec{k}}{2}) c_{\vec{k} + \frac{\vec{k}}{2}}^\dagger c_{\vec{k}} \right) =$$

$$= -\frac{e}{m} \sum_{\vec{k}} \left(\vec{k} + \frac{\vec{k}}{2} \right) \cdot \psi_{\vec{k} + \frac{\vec{k}}{2}}^\dagger \psi_{\vec{k}}$$

$$\rightarrow H_{cr}^{(1)} = \frac{e}{m} \sum_{\vec{k} \in \vec{a}} \left(\vec{k} + \frac{\vec{k}}{2} \right) \cdot \psi_{\vec{k} + \frac{\vec{k}}{2}}^\dagger \psi_{\vec{k}} (a_{\frac{\vec{k}}{2},1} + a_{-\frac{\vec{k}}{2},1}^\dagger) \frac{1}{\sqrt{2V\epsilon_0 \omega_{\vec{k},1}}}$$

Electron-photon red $\mathcal{O}(e^2)$!

$$H_e^{(0)} = \sum_{\delta, j} \varepsilon_{\delta}^{(0)} c_{\delta j}^{\dagger} c_{\delta j} - \sum_{j, \delta, \delta'} t_{\delta} (c_{\delta j}^{\dagger} c_{\delta' j + \vec{\delta}} + \text{h.c.}) \rightarrow \varepsilon_{\delta j} = \varepsilon_{\delta}^{(0)} - 2t_{\delta} (\cos k_x + \cos k_y)$$

$$t_{\delta} \rightarrow t_{\delta} e^{i \theta_{\delta j}} \quad \theta_{\delta j} = e \int_{\vec{R}_j}^{\vec{R}_{j+\vec{\delta}}} \vec{A}(\vec{r}, t) \cdot d\vec{r}$$

$$H_e^{(0)}[A] = \sum_{\delta, j} \varepsilon_{\delta}^{(0)} c_{\delta j}^{\dagger} c_{\delta j} - \sum_{j, \delta, \delta'} t_{\delta} (e^{i \theta_{\delta j}} c_{\delta j}^{\dagger} c_{\delta' j + \vec{\delta}} + e^{-i \theta_{\delta' j}} c_{\delta' j + \vec{\delta}}^{\dagger} c_{\delta j})$$

$$e^{i\theta} = 1 + i\theta - \frac{1}{2}\theta^2$$

$\mathcal{O}(eA) \quad \mathcal{O}(eA^2)$

$$c_{\delta j} = \frac{1}{\sqrt{V}} \sum_{\vec{k}} e^{i \vec{k} \cdot \vec{R}_j} c_{\delta \vec{k}}$$

$$H_{e\gamma}^{(1)} = - \sum_{j, \delta, \delta'} t_{\delta} \left(\frac{e^2}{2} \right) \cdot (c_{\delta j}^{\dagger} c_{\delta' j + \vec{\delta}} - c_{\delta' j + \vec{\delta}}^{\dagger} c_{\delta j}) = \approx \vec{A}(\vec{R}_j + \frac{\vec{\delta}}{2}) \cdot \vec{\delta}$$

$(\vec{A} \text{ slowly varying})$

$$= \sum_{j, \delta, \delta'} t_{\delta} \cdot \frac{e^2}{2} \int_{\vec{R}_j}^{\vec{R}_{j+\vec{\delta}}} \vec{A}(\vec{r}, t) d\vec{r} \int_{\vec{R}_j}^{\vec{R}_{j+\vec{\delta}}} \vec{A}(\vec{r}, t) d\vec{r} (c_{\delta j}^{\dagger} c_{\delta' j + \vec{\delta}} - \text{h.c.}) =$$

$$= \sum_{j, \delta, \delta'} t_{\delta} \frac{e^2}{2} \left(\sum_{\vec{q}, \vec{q}'} \frac{1}{\sqrt{2\varepsilon_0 V \omega_{\vec{q}}}} (\vec{C}_{\vec{q}} \cdot \vec{\delta}) e^{i \vec{q} \cdot (\vec{R}_j + \frac{\vec{\delta}}{2})} (b_{\vec{q}, \delta}^{\dagger} + b_{-\vec{q}, \delta}^{\dagger}) \right) \left(\sum_{\vec{q}', \vec{q}'} \frac{1}{\sqrt{2\varepsilon_0 V \omega_{\vec{q}'}}} (\vec{C}_{\vec{q}'} \cdot \vec{\delta}) e^{i \vec{q}' \cdot (\vec{R}_j + \frac{\vec{\delta}}{2})} (b_{\vec{q}', \delta'}^{\dagger} + b_{-\vec{q}', \delta'}^{\dagger}) \right)$$

$$= \sum_{\vec{k}, \vec{k}'} \frac{1}{V} c_{\delta \vec{k}}^{\dagger} c_{\delta' \vec{k}'} e^{-i(\vec{k}' - \vec{k}) \cdot \vec{R}_j} \underbrace{\left(e^{i \vec{k} \cdot \vec{\delta}} - e^{-i \vec{k}' \cdot \vec{\delta}} \right)}_{2i \sin(\vec{k} \cdot \vec{\delta})} =$$

$$\sum_{\vec{k}} c_{\delta \vec{k} + \frac{\vec{\delta}}{2}}^{\dagger} c_{\delta' \vec{k}}$$

$$\sum_{\vec{q}} \frac{1}{V} e^{i(\frac{\vec{\delta}}{2} + \frac{\vec{\delta}}{2} - \vec{k}' + \vec{k}) \cdot \vec{R}_j} = \delta_{\vec{k}', \vec{k} + \frac{\vec{\delta}}{2}}$$

$$c_{\vec{j}}^{\dagger} c_{\vec{j}+\vec{\delta}} - c_{\vec{j}+\vec{\delta}}^{\dagger} c_{\vec{j}} = \frac{1}{V} \sum_{\vec{k}, \vec{k}'} c_{\vec{k}}^{\dagger} c_{\vec{k}'} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} (e^{i \vec{k} \cdot \vec{\delta}} - e^{-i \vec{k}' \cdot \vec{\delta}})$$

$$\frac{1}{V} \left(\sum_{\vec{k}'} e^{i \vec{k}' \cdot \vec{r}} c_{\vec{k}'}^{\dagger} \sum_{\vec{k}} e^{i \vec{k} \cdot \vec{r}} e^{i \vec{k} \cdot \vec{\delta}} c_{\vec{k}} \right)^{\dagger} =$$

$$= \frac{1}{V} \sum_{\vec{k}, \vec{k}'} c_{\vec{k}}^{\dagger} c_{\vec{k}'} e^{i \vec{k} \cdot \vec{r}} e^{-i \vec{k}' \cdot \vec{r}} e^{i \vec{k} \cdot \vec{\delta}} = \frac{1}{V} \sum_{\vec{k}(\vec{k} < \vec{k}')} c_{\vec{k}}^{\dagger} c_{\vec{k}'} e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} e^{-i \vec{k}' \cdot \vec{\delta}}$$

